

Week 5 — Experiments, Results & Figures Complete

The final week collects every result and produces every figure and table for BOTH articles. By Day there is no coding or experimentation left — only manuscript assembly, which continues afterward. Team A produces the headline guarantee figures and all statistics; Team B produces the detection, robustness and disentanglement figures; Team C finalises Article-2 assets and cross-checks the shared feature-selection story.

Team A — Coverage Figure, Zero-Day Tables & Final Statistics

Members: Interns 1, 2, 3 (Amon Koike, Mohammed Owais, Iwo Wojtakajtis)

Focus this week: Deliver the paper's headline guarantee evidence: the coverage plot and the complete, significance-tested result tables.

Day	Tasks	Deliverables
Day 26	Support the disentanglement experiment: supply the adversarial-known vs true-zero-day labels and scoring. Compute separation AUROC between the two populations.	Disentanglement scoring Separation AUROC v1
	Finalise Table A (in-distribution detection) with significance markers across all datasets. Cross-validate coverage across all seeds.	Table A (final numbers) All-seed coverage check
	Freeze all Team-A result artefacts for the experiment/figure week. Confirm every RQ2/RQ5 number is reproducible from the frozen pipeline.	Frozen Team-A results Reproducibility confirmation
Day 27	Generate Figure 2 (headline): empirical false-zero-day rate vs target α — must track the diagonal across seeds and datasets. Add per-dataset overlays and confidence bands.	Figure 2 — coverage plot (final)
Day 28	Finalise Table B (zero-day): true-zero-day recall + achieved α vs target, all datasets, with the guarantee visibly holding. Re-run coverage across all 5 seeds for the CI.	Table B (final) All-seed coverage CIs
Day 29	Finalise Table A (in-distribution detection) with significance markers vs every baseline. Attach Cohen's d effect sizes.	Table A (final, with effects)
	Produce the conformal-vs-heuristic ablation figure/table (shows the guarantee's value). Compile the RQ5 honesty summary: where quantum helps / does not.	Conformal ablation asset RQ5 honesty summary
Day 30	Cross-audit all Team-A numbers against raw logs; fix any rounding/label mismatches. Hand final statistics + coverage assets to Team B for manuscript assembly.	Audited statistics Handover to manuscript
Day 31	Write the Data + Conformal-Statistics methods subsections (partition protocol, calibration, coverage). Prepare the reproducibility appendix for the statistics pipeline.	Methods subsections (data + stats) Repro appendix (stats)
Day 32	Final sign-off: every RQ2 / RQ5 number and Figure 2 reproducible from one script. Confirm all Team-A deliverables are frozen for writing.	✅ All Team-A results & figures COMPLETE Repro sign-off

Team B — Detection, Robustness & Disentanglement Figures + Article-1 Assembly

Members: Interns 4, 5 (Arjun E Naik, La Wun Nannda)

Focus this week: Produce the trust-region, disentanglement and robustness figures and assemble the Article-1 manuscript around the theory and results.

Day	Tasks	Deliverables
Day 26	FULL INTEGRATION run across all datasets; hand separation scores to Team A for AUROC. Collect certified-accuracy-at-radius curves (RQ4).	Full-integration results Certified-accuracy curves
	Run the MAQT ablation (CE-only vs MAQT) to isolate the geometry-shaping benefit. Run the noise-rate p ablation (certified vs empirical).	MAQT ablation results Noise-rate ablation results
	Freeze all QS-Net model artefacts and raw experiment outputs. Confirm one-command re-run of the whole inference pipeline.	Frozen QS-Net artefacts One-command rerun check
Day 27	Generate Figure 1: trust-region visualisation (2D projection of Hilbert-space prototypes + samples). Generate Figure 3: disentanglement — separation AUROC vs attack budget with the Proposition-2 threshold marked.	Figure 1 — trust regions Figure 3 — disentanglement
Day 28	Finalise Table C (robustness): certified + empirical accuracy under FGSM/PGD across ϵ and noise p. Plot certified-radius distributions.	Table C (final) Certified-radius figure
Day 29	Finalise the MAQT ablation and noise-rate ablation figures. Confirm H1 (tighter intra / larger inter fidelity gaps) with final numbers.	Ablation figures H1 confirmation
Day 30	Assemble Article-1 Methodology + Theory sections (Algorithms 1–3, Propositions 1–3) from the methodology document. Insert the finalised RQ1/RQ3/RQ4 results.	Article-1 Methodology + Theory Results inserted
	Integrate Team A's coverage figure + zero-day tables and Team C's feature-selection link into Article 1. Draft the Article-1 Discussion (incl. honest negatives).	Integrated Article-1 draft Discussion draft
Day 31	Assemble the Dockerised repo + one-command rerun for Article 1 (reproducibility strength). Draft Abstract + Introduction from the methodology's title/abstract sketch.	Repro package (Article 1) Abstract + Intro draft
Day 32	Final experiment/figure sign-off for Article 1: all figures (1,2,3) + tables (A,B,C) present and reproducible. Freeze the Article-1 manuscript skeleton for post-Day-35 polishing.	✅ All Article-1 figures/tables COMPLETE Frozen manuscript skeleton

Team C — Article-2 Experiments to Completion, and Article-2 Finalisation & Cross-Article Feature Story

Members: Interns 6, 7 (Kevin Wong, Maral Mahmoudi Kamelabad)

Focus this week: *Finish every Article-2 experiment and lock its results, so only writing remains.*

Focus this week: *Lock every Article-2 asset and make sure the feature-selection narrative is consistent across both papers. Leave it the paper writing part*

Day	Tasks	Deliverables
Day 26	Run the complete selector benchmark (accuracy / #features / runtime / stability) with final settings, all datasets, 5 seeds. Apply Holm–Bonferroni across selector comparisons.	Final selector benchmark (locked) Corrected significance
	Finalise the selector → Zero-Day AUROC experiment across all datasets. Add MO-AQPSO Pareto-front analysis (accuracy vs feature count).	Selector→AUROC (final) Pareto-front analysis
Day 27	Measure computational cost (runtime, evaluations) for each selector for the cost table. Cross-check MO-AQPSO subsets are the ones used by QS-Net (consistency with Article 1).	Cost comparison table Cross-article consistency check
	Generate all Article-2 figures at publication resolution (comparison bars, Pareto front, AUROC-by-selector). Assemble the two headline Article-2 tables in final form.	Article-2 figures (final) Article-2 tables (final)
Day 28	Write Article-2 Results + Discussion around the locked numbers. Draft the Article-2 Conclusion and limitations.	Article-2 Results + Discussion Article-2 Conclusion draft
	Full internal read-through of Article 2 with mentors; incorporate fixes. Verify every figure/table is reproducible from a script.	Reviewed Article-2 manuscript Reproducibility check (Article 2)
Day 29	Freeze all Article-2 experimental artefacts and figures. Prepare Article-2 for the final polishing that continues past Day 35. Finalise the two Article-2 headline tables (selector comparison; selector → zero-day AUROC).	Frozen Article-2 results Article-2 near-final draft
Day 30	Regenerate all figures at final resolution.	Deliverables Article-2 tables (locked) Article-2 figures (locked)
	Verify MO-AQPSO is the recommended selector with evidence; write the recommendation subsection. Add stability + cost discussion.	Recommendation subsection Stability/cost discussion
Day 31	Ensure the feature subset used in QS-Net (Article 1) exactly matches the Article-2 winner; document the linkage. Write the cross-reference paragraph used in both papers.	Cross-article consistency doc Shared cross-reference text
	Complete Article-2 Results + Discussion with final numbers and significance. Finalise limitations + future work.	Article-2 Results/Discussion (final) Limitations + future work
Day 32	Mentor review pass on the full Article-2 manuscript; apply edits. Verify every number traces to a script/log.	Reviewed Article-2 Traceability check
	Assemble the Article-2 reproducibility package (repo + environment + rerun script). Prepare submission-format version (target venue template).	Repro package (Article 2) Formatted Article-2 draft
	Final sign-off: all Article-2 experiments, figures and tables COMPLETE and reproducible. Hand the shared feature-selection assets to Team B for Article 1.	✓ All Article-2 results/figures COMPLETE Shared assets handed over

